

V1 VS V2

The conclusion: Across two independent 1,000-user offline evaluations, SVD consistently outperformed the standalone 3-agent recommender, confirming that behavioural collaborative signals remain important for e-commerce recommendation. The Hybrid SVD + agentic reranking framework consistently and substantially outperformed the standalone 3-agent method, showing that the agentic component is more effective as an augmentation layer than as an independent recommender. Compared with SVD, the Hybrid method remained competitive in HitRate@10 and showed stronger NDCG@10 in the seed99 robustness run, but it did not consistently demonstrate a statistically clear HitRate@10 advantage and it consistently reduced intra-list diversity. Therefore, the Hybrid framework should be positioned as a reranking and explainability-oriented augmentation of SVD, rather than a full replacement.

Method	Run 1 HitRate@10	Seed99 HitRate@10	Run 1 NDCG@10	Seed99 NDCG@10	Run 1 ILD@10	Seed99 ILD@10
SVD	0.500000	0.505000	0.295493	0.296245	0.751933	0.755089
Standalone 3-Agent	0.274000	0.269000	0.155365	0.154352	0.543800	0.539319
Hybrid	0.498000	0.526000	0.304552	0.312443	0.694481	0.692919

Experiment V2 results:

Seed99 Robustness Experiment Report

Repo Context

- Repo path: `D:\GOOGLE DRIVE\EMERALD PATHWAYS\WEB WORK\AI CODING\VS CODE\personal\LILI FINAL PAPER DEMO`
- Branch: `formal-experiment-seed99-robustness`
- Latest commit at run time: `a6177c0 Update formal experiment API UI and handoff`
- Previous safety commit: `8d965b1 Add safe seed99 artifact naming for formal experiment`

Run Configuration

- Artifact prefix: `seed99\_robustness`
- Random seed: `99`
- Subset size: `1000` valid users
- Evaluation protocol: leave-one-out

- Candidate pool size: `100` items per user
- Candidate pool rule: held-out ground-truth item included for every user
- Shared candidate pool: yes, reused across SVD, standalone 3-agent, and Hybrid
- Recommendation cutoff: Top-10
- Metrics: HitRate@10, NDCG@10, ILD@10
- Uncertainty estimation: bootstrap 95% confidence intervals with `1000` samples

Methods

1. SVD Matrix Factorisation
2. Standalone 3-Agent recommender
3. Hybrid SVD + 3-Agent reranker

Hybrid formula used in this run:

```
``text
hybrid_score = 0.70 * normalized_svd_score
               + 0.25 * normalized_agentic_score
               + 0.05 * diversity_bonus
...

```

Validation Checks

- `users\_evaluated = 1000`
- SVD recommendations for 1000 users: yes
- Standalone 3-Agent recommendations for 1000 users: yes
- Hybrid recommendations for 1000 users: yes
- Full Top-10 produced for every user and every method: yes
- All Top-10 recommendations inside each user's candidate pool: yes
- No Top-10 recommendation contains training items: yes
- Ground-truth item inside every candidate pool: yes
- Duplicate candidate items in any candidate pool: no
- `article\_id` and `customer\_id` treated as strings: yes
- Metric calculation errors: `[]`
- Missing metadata for diversity: `0`
- Internal consistency issues: `0`
- Bootstrap CI file generated: yes

Seed99 Metric Results

SVD Matrix Factorisation

- HitRate@10: `0.505000`
- NDCG@10: `0.296245`
- ILD@10: `0.755089`

Standalone 3-Agent

- HitRate@10: `0.269000`
- NDCG@10: `0.154352`
- ILD@10: `0.539319`

Hybrid SVD + 3-Agent Reranker

- HitRate@10: `0.526000`
- NDCG@10: `0.312443`
- ILD@10: `0.692919`

Bootstrap 95% Confidence Intervals

Per-Method

- SVD HitRate@10: mean `0.504529`, 95% CI `[0.472975, 0.536000]`
- SVD NDCG@10: mean `0.296098`, 95% CI `[0.274121, 0.317381]`
- SVD ILD@10: mean `0.755005`, 95% CI `[0.750296, 0.759726]`
- Standalone 3-Agent HitRate@10: mean `0.269383`, 95% CI `[0.243000, 0.296000]`
- Standalone 3-Agent NDCG@10: mean `0.154547`, 95% CI `[0.137662, 0.173458]`
- Standalone 3-Agent ILD@10: mean `0.539313`, 95% CI `[0.533229, 0.545904]`
- Hybrid HitRate@10: mean `0.525113`, 95% CI `[0.496000, 0.556000]`
- Hybrid NDCG@10: mean `0.312236`, 95% CI `[0.290299, 0.333084]`
- Hybrid ILD@10: mean `0.692825`, 95% CI `[0.687242, 0.698304]`

Pairwise Differences

- Hybrid minus SVD HitRate@10: mean `0.020584`, 95% CI `[-0.001000, 0.042000]`
- Hybrid minus SVD NDCG@10: mean `0.016139`, 95% CI `[0.007190, 0.025000]`
- Hybrid minus SVD ILD@10: mean `-0.062179`, 95% CI `[-0.066816, -0.057555]`
- Hybrid minus Standalone 3-Agent HitRate@10: mean `0.255730`, 95% CI `[0.225950, 0.286000]`
- Hybrid minus Standalone 3-Agent NDCG@10: mean `0.157690`, 95% CI `[0.133920, 0.180924]`
- Hybrid minus Standalone 3-Agent ILD@10: mean `0.153512`, 95% CI `[0.146976, 0.160104]`

Comparison With Previous Non-Prefixed 1,000-User Run

Previous non-prefixed results:

- SVD: HitRate@10 `0.500000`, NDCG@10 `0.295493`, ILD@10 `0.751933`
- Standalone 3-Agent: HitRate@10 `0.274000`, NDCG@10 `0.155365`, ILD@10 `0.543800`
- Hybrid: HitRate@10 `0.498000`, NDCG@10 `0.304552`, ILD@10 `0.694481`

Seed99 directional comparison:

- SVD remains stronger than standalone 3-agent on HitRate@10 and NDCG@10.
- Hybrid improves over standalone 3-agent on HitRate@10, NDCG@10, and ILD@10.
- Hybrid does not clearly beat SVD on HitRate@10 because the Hybrid-minus-SVD HitRate CI crosses zero.

- Hybrid does exceed SVD on NDCG@10 in this seed99 run because the Hybrid-minus-SVD NDCG CI stays above zero.
- Hybrid remains less diverse than SVD because the Hybrid-minus-SVD ILD CI stays below zero.
- The overall direction is broadly consistent with the previous 1,000-user run: SVD stays much stronger than standalone 3-agent, Hybrid stays much stronger than standalone 3-agent, and Hybrid remains less diverse than SVD.

Careful Interpretation

- This seed99 robustness run supports the same broad qualitative story as the previous 1,000-user sample.
- SVD remains the strongest standalone offline baseline versus the standalone 3-agent method.
- Hybrid appears to improve ranking quality over standalone 3-agent by a wide margin.
- Hybrid shows higher NDCG@10 than SVD in this seed99 run, but it does not show a clear HitRate@10 win over SVD because that confidence interval includes zero.
- Hybrid remains less diverse than SVD, not more diverse.
- These are offline ranking results only. They do not establish live customer engagement, CTR, CVR, or validated feedback adaptation.

Limitations

- Single dataset, offline evaluation only.
- Fixed Hybrid heuristic; no tuning was performed in this run.
- Robustness here means a second random 1,000-user sample, not universal generalisation.
- Confidence intervals reflect bootstrap uncertainty for this offline setup only.

Experiment V1 results:

Expanding to 1,000-user results:

Method	HitRate@10	NDCG@10	ILD@10
SVD	0.500	0.295493	0.751933
3-Agent	0.274	0.155365	0.543800

SVD significantly outperforms standalone 3-agent, and bootstrap CI shows a stable difference, not random fluctuations. The diagnostic report suggests standalone 3-agent is too concentrated in metadata attributes and does not generalise as well as SVD's behavioural signal.

Given that standalone 3-agent underperforms SVD, is it academically acceptable to reposition the agentic framework as an augmentation/reranking layer over SVD, rather than as a full replacement for CF? (**the first thing that needs to be confirmed by you**)

And I compared the 3 methods:

Method	HitRate@10	NDCG@10	ILD@10
SVD	0.500	0.295493	0.751933
Standalone 3-Agent	0.274	0.155365	0.543800
Hybrid	0.498	0.304552	0.694481

Hybrid substantially improves over standalone 3-agent and reaches SVD-comparable relevance, but SVD remains the strongest baseline overall.

Should I refine the research framing from “agentic AI outperforming CF” to “agentic AI augmenting CF through explainability, evidence-based reranking and controllability”? (**the second thing that needs to be confirmed by you**)

Should I keep the standalone 3-agent result as an ablation study to show that the agentic layer alone is weaker than SVD, but becomes more useful when combined with SVD? (**the third thing that needs to be confirmed by you**)